Fiscal Year:	FY 2024	**Task Last Updated:**	FY 01/12/2024
PI Name:	Hammond, Timothy M.D.		
Project Title:	Fuel to Mars		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Microbiology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	(1) Translational (Countermeasure) Potential		
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Comments:	Alternate email: Timothy.Hammond@va.gov NOTE: PI previously at Tulane University until 2007 (Ed., 09/2010) Joint appointments Professor, Department of Internal Medicine, Nephrology Division, Duke University School of Medicine, and Staff Physician, Durham VA Health Care System.		
Project Type:	Flight	**Solicitation / Funding Source:**	2018 Space Biology (ROSBio) NNH18ZTT001N-Artemis1 (EM1). App A: Orion (Artemis-1) (formerly Exploration Mission-1)
Start Date:	03/18/2019	**End Date:**	03/17/2024
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	1
No. of Master's Candidates:	3	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA KSC
Contact Monitor:	Zhang, Ye	**Contact Phone:**	321-861-3253
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 03/17/2024 per NSSC information (Ed., 1/4/24). NOTE: End date changed to 03/17/2023 per NSSC information (Ed., 9/22/22).		
Key Personnel Changes/Previous PI:	January 2021 report: None.		
COI Name (Institution):	Allen, Patricia M.S. (Veterans Health Administration) Birdsall, Holly M.D., Ph.D. (Institute For Medical Research, Inc.) Giaever, Guri Ph.D. (University of British Columbia, Canada) Nislow, Corey Ph.D. (University of British Columbia, Canada)		
Grant/Contract No.:	80NSSC19K0706		

Performance Goal No.:	
Performance Goal Text:	NOTE: End date changed to 3/17/2022 per NSSC information (Ed., 5/18/21)
Task Description:	The primary research product this project will produce is identification of the insertion strains of Chlamydomonas reinhardtii, a single-cell green alga, which optimizes survival during exposure to the combined impacts of space radiation and microgravity in the cosmic space environment beyond the Van Allen Belts. The Chlamydomonas reinhardtii strains in a mapped insertional mutant library are a transformative tool allowing near genome wide analysis of the genes mediating survival advantage during stimuli such as cosmic radiation and microgravity. The secondary set of research products this project will produce are identification of the pathways mediating the gene changes, visualization of the pathway interactions, and identification of any environments giving similar gene pathway signatures in other model systems. Our goal is to identify strains of Chlamydomonas reinhardtii with gene insertions, which convey the best survival advantage during the exposure to the combined impacts of space radiation and microgravity for future use as parent strains in hydrogen production initiatives. Hydrogen is the lead contender for fuel production for return space flight missions from orbital bodies beyond low Earth orbit. The metabolism of select microorganisms makes hydrogen fuel. The best-known and most efficient hydrogen producer is Chlamydomonas reinhardtii, which is poised for exploitation as an industrial biotechnology platform for biofuel production. The type of investigation we propose is experiments that will be conducted aboard the Exploration Mission-1 (EM-1) [Note now called Artemis] lunar orbit flight, with asynchronous controls in Kennedy Space Center ISSES (International Space Station Environmental Simulators) Chambers. The proposed hardware is the NASA Biological Research in Canisters (BRIC)-100VC. Hypothesis: The hypothesis to be tested is that the mapped insertional mutant library of Chlamydomonas reinhardtii will allow identification of the genes that mediate optimal radiation resistance, and imperviousness to microgravity, in the cosmic space environment beyond the Van Allen Belts. The specific aims of the proposal are: Specific Aim 1: To experimentally determine which of the thousands of mapped insertion mutant strains of Chlamydomonas reinhardtii have the best long-term survival advantage during combined space radiation and microgravity impacts in the cosmic space environment beyond the Van Allen Belts. Specific Aim 2: To analyze the dataset of insertion strains with the best long-term survival advantage during combined space radiation and microgravity impacts to: • Perform pathway analysis using gene set enrichment analysis (GSEA); • Visualize the pathway interactions using Cytoscape/ClueGo; • Make comparison to homologous gene sets in Chlamydomonas, Saccharomyces cerevisiae, and other model organism databases for any environments giving similar gene pathway signatures. Although focused on gene pathways for survival for future applications in hydrogen fuel production, the data generated will provide genome level insights into the cancer risks of prolonged exposure to cosmic radiation and microgravity. The spectrum of cancer risk genes is known, and this study will parse the importance of risk genes, and the patterns of radiation risk change with exposure to the cosmic space environment beyond the Van Allen Belts.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The research impacts and Earth benefits include: o The first opportunity to apply modern “omics” techniques to samples flown beyond the van Allen belts, and recover for ground-based analysis. o The first biological validation of the utility of the Chlamydomonas deletion series for academic and commercial applications. o Delineation of the production of the potential fuels, lipid and hydrogen, to optimize use in future space based and ground applications. o Education of the next generation of space flight investigators. o Identifies mechanisms of radiation damage due to cosmic radiation with possible future clinical applications. o This proposal directly addresses the Space Biology Plan Element GL3: What are the key cellular and molecular systems necessary for biological organisms to thrive in space environment that can be discovered from 21st century bioinformatics tools? o This application directly addresses three elements of the Human Research Roadmap on Risk of Radiation Carcinogenesis: Cancer 07: How can systems biology approaches be used to integrate research on the molecular, cellular, and tissue mechanisms of radiation damage to improve the prediction of the risk of cancer and to evaluate the effectiveness of countermeasures? How can epidemiology data and scaling factors support this approach? Cancer 08: What are the most effective medical or dietary countermeasures to mitigate cancer risks from exposure to SPE (solar particle events) and GCR (galactic cosmic rays)? What side effects should be tolerated versus mission risks? Can medical countermeasures that provide mitigation across multiple cancer types or across multiple radiation risk areas be identified? Cancer 09: Are there significant effects from other spaceflight factors that modify the carcinogenic risk from space radiation?

Task Progress:

Artemis I flight experiment:
The Space Launch System (SLS) rocket launched November 16, 2022 from Launch Complex 39B at the NASA Kennedy Space Center. The Orion spacecraft was aboard the Artemis-I rocket, together with the Fuel to Mars experiment in the new Moonshot hardware, for a planned 25.5-day space mission. After quickly reaching Earth orbit, the upper stage separated and performed a trans-lunar injection (burn to the Moon). The Orion spacecraft completed one flyby of the Moon on November 21, entered a distant retrograde orbit for six days and completed a second flyby of the Moon on November 25, before returning home to Earth on December 11, 2022. The Orion capsule was recovered from the Pacific Ocean, by the U.S.S. Portland, taken into San Diego Bay, placed on a truck and driven to Kennedy Space Center. The Space Biology experiments were salvaged from Orion and released to the investigators shortly after arrival at Kennedy Space Center.

The Chlamydomonas mapped insertional mutant library was maintained until launch on Tris-Acetate-Phosphate (TAP) agar plates and aliquots reseeded onto fresh plates robotically every 2 weeks. There have been no growth or contamination problems to date over a period beyond 24 months, including prior to launch on the pad.

Both the flight and ground control copies of the Moonshot hardware performed flawlessly during the flight. The hardware performed nominally throughout the verification flight. The maximum temperature change was 12°C, well within the envelope of tolerable experimental conditions. At the end of the ~25.5-days flight, the batteries still had more than 84% of power remaining. The power profile validated that the lights came on for 6 out of 24 h daily, as planned.

Acceleration recordings showed serial measurements of gravity in the z axis, but no acceleration in other directions as expected.

The Moonshot hardware is flight certified and flight proven. The ground control measurements from the sensors in the hardware were no different between flight and ground, except that the temperature variation was less marked in the ground controls.

Cost and practicality of Moonshot hardware:

Advances in materials, microelectronics, and molecular and cellular biology technologies provide a firm basis to produce cost-effective customized spaceflight hardware. Following this paradigm, we report the design, construction, and validation of a simple cost-effective flight hardware for growth of the green algae C. reinhardtii. The new hardware, termed “Moonshot,” flew in the Orion capsule of the first Artemis mission, a circumnavigation of the moon.

Both copies of the flight hardware, and a prototype were designed, manufactured, and validated for a total cost for all components combined, less than 10% of the cost of similar hardware by other groups.

Initial analysis of the samples on recovery post flight unexpectedly showed better survival of Chlamydomonas in flight during cosmic radiation exposure than on the ground. Four lines of evidence indicate possible mechanisms for these findings.

- First, there is no difference in energy metabolism via chlorophyll or photosystem pathways between flight and ground samples. Specifically, there is no difference in Chlorophyll A and B, Photosystem I and II, and Cytochrome f are no different between flight and ground samples. However, lipid content of cells is significantly reduced in flight compared to ground samples.

- Second, there are no differences in Chlamydomonas population growth between flight and ground samples. Growth curves by serial spectrophotometer measures of 800nM, growth of aliquots samples in TAP, movies of Chlamydomonas swimming, and flow cytometry analysis of live cells, all prove that some live Chlamydomonas are present on sample recovery.

- Third, there are changes in programmed cell death pathways during flight. Flight samples showed significantly more apoptosis, and significantly less cell death by necrosis, than ground controls. The apoptotic cells were larger than the necrotic cells during flow cytometry analysis.

- Fourth, as a discovery level investigation of molecular mechanisms mediating differences between flight and ground samples were imaged by fluorescence and Ramen spectroscopy. Fluorescence imaging confirms two cell populations with different cell size in the flight samples, as shown earlier in the flow cytometry analysis. The larger cells are much more prevalent in the flight than ground samples. By Raman spectroscopy, the larger cells have significantly increased beta-carotene compared to the smaller flight cells, ground control cells, or azide killed control cells. Beta-carotene is a key component of the cell redox system.

- Last, aliquots of Chlamydomonas had the tardigrade D-Sup gene, known to be protective during diverse ground-based radiation exposures, were transfected into their nuclei. A green fluorescent protein (GFP) marker was included in the D-Sup cassette. Nuclear D-sup localization and expression was confirmed by fluorescent microscopy. The D-Sup transfected Chlamydomonas had better survival both in flight and on the ground compared to irrelevant control transfected strains.

Ongoing analysis is experimentally determining which of the thousands of mapped insertion mutant strains of Chlamydomonas reinhardtii have the best long-term survival advantage during combined space radiation and microgravity impacts in the cosmic space environment beyond the Van Allen Belts.

DNA has been extracted from all flight and ground control samples. DNA integrity and quality of all samples has been validated. Rapid Amplification of cDNA Ends (RACE) has been performed on all samples, and the resultant products sequenced. Analysis of genes providing a survival advantage during exposure to cosmic radiation is in progress. Pathway analysis and comparison to homologous gene sets will follow.

All success criteria have been met at the superior level to progress to address the primary goals of the completed aims.

Bibliography Type: Description: (Last Updated: 03/31/2026)

Abstracts for Journals and Proceedings	Hammond T, Allen P, Gaikani HK, Nislow C, Giaever G, Russick JM, Wells H, Panikkanvalappil S, Levine H, Zhang Y, Birdsall H. "Protecting Chlamydomonas reinhardtii from cosmic radiation on the Artemis I mission." 39th Annual Meeting of the American Society for Gravitational and Space Research, Washington, DC, November 13-18, 2023. Abstracts. 39th Annual Meeting of the American Society for Gravitational and Space Research, Washington, DC, November 13-18, 2023. , Nov-2023
Abstracts for Journals and Proceedings	Hammond T, Allen P, Gaikani HK, Nislow C, Giaever G, Russick JM, Wells H, Panikkanvalappil S, Zhang Y, Birdsall H. "Fuel to Mars (Artemis I Biological Payloads General Session)." 39th Annual Meeting of the American Society for Gravitational and Space Research, Washington, DC, November 13-18, 2023. Abstracts. 39th Annual Meeting of the American Society for Gravitational and Space Research, Washington, DC, November 13-18, 2023. , Nov-2023